

DAY 1 (Variables, Data Types, I/O, Operators)

1.

Write a program to input two numbers and print their sum.

2.

Write a program to calculate area of a rectangle.

3.

Write a program to calculate simple interest.

4.

Write a program to convert Celsius into Fahrenheit.

5.

Write a program to input employee basic salary and calculate HRA (20%) and DA (10%).

DAY 2 (Conditions)

6.

Write a program to check whether a number is positive, negative or zero.

7.

Write a program to find the largest among two numbers.

8.

Write a program to find the largest among three numbers.

9.

Write a program to check whether a number is even or odd.

10.

Write a program to check whether a year is leap year or not.

DAY 3 (Loops)

11.

Write a program to print numbers from 1 to 100.

12.

Write a program to print even numbers between 1 and 100.

13.

Write a program to find sum of first N natural numbers.

14.

Write a program to print multiplication table of a given number.

15.

Write a program to calculate factorial of a number.

DAY 4 (Functions)

16.

Write a function to find square of a number.

17.

Write a function to find maximum of two numbers.

18.

Write a function to calculate simple interest.

19.

Write a function to check whether a number is even or odd.

20.

Write a function to calculate area of a circle.

DAY 5 (Lists, Tuples, Dictionary, Set)

21.

Write a program to create a list of 10 numbers and print all elements.

22.

Write a program to find largest element in a list.

23.

Write a program to calculate sum of all elements in a list.

24.

Write a program to count even numbers in a list.

25.

Write a program to remove duplicate elements using set.

10 Mixed Questions

26.

Write a program to reverse a number.

27.

Write a program to check whether a number is palindrome.

28.

Write a program to count digits in a number.

29.

Write a program to find sum of digits of a number.

30.

Write a program to generate Fibonacci series up to N terms.

31.

Write a program to check whether a number is prime.

32.

Write a program to print all prime numbers between 1 and 100.

33.

Write a program to count vowels in a string.

34.

Write a program to reverse a string.

35.

Write a program to count frequency of each character in a string.

1. File Creation

Question: Create a file `student.txt` and store 5 student names entered by the user.

Hint:

- Use `open("student.txt", "w")`
 - Use a loop to take 5 inputs.
 - Use `write()` and `"\n"`.
-

2. Read Entire File

Question: Read and display complete contents of `student.txt`.

Hint:

- Open file in read mode.
 - Use `read()`.
 - Print the returned string.
-

3. Count Characters

Question: Count total characters present in a file.

Hint:

- Read entire file using `read()`.
 - Use `len(data)`.
-

4. Count Lines

Question: Count total number of lines in a file.

Hint:

- Use `readlines()`.

- Length of list = total lines.
-

5. Count Words

Question: Count total words in a file.

Hint:

- Read complete file.
 - Use `split()`.
 - Count length of resulting list.
-

6. Read Line by Line

Question: Display file contents line by line using `readline()`.

Hint:

- Use a loop.
 - Stop when `readline()` returns empty string.
-

7. Append Data

Question: Add 3 new records to an existing file without deleting old data.

Hint:

- Open file in append mode (`a`).
 - Use `write()`.
-

8. Search a Word

Question: Check whether a given word exists in a file.

Hint:

- Read file content.
- Take word from user.

- Use `if word in data`.
-

9. Count Vowels

Question: Count total vowels present in a file.

Hint:

- Read file content.
 - Traverse character by character.
 - Check against `"aeiouAEIOU"`.
-

10. Count Uppercase and Lowercase Letters

Question: Count uppercase and lowercase characters separately.

Hint:

- Use `isupper()` and `islower()`.
 - Maintain two counters.
-

11. Display Longest Line

Question: Find the longest line in a text file.

Hint:

- Use `readlines()`.
 - Compare lengths of lines.
 - Store maximum line.
-

12. Copy File Content

Question: Copy content of one file to another.

Hint:

- Read source file.

- Write content into destination file.
-

13. Replace a Word

Question: Replace all occurrences of "Python" with "Programming".

Hint:

- Use `replace()`.
 - Write modified content back to file.
-

14. Store and Read Marks

Question: Store marks of 10 students and display students scoring more than 75.

Hint:

- Store as:

```
Rahul,82  
Aman,67
```
 - Read line by line.
 - Use `split(",")`.
-

15. Binary File (Employee Records)

Question: Store employee records in a binary file and search by EmpID.

Hint:

- Import `pickle`.
 - Use `pickle.dump()` for writing.
 - Use `pickle.load()` for reading.
 - Compare entered EmpID with stored records.
-

Advanced Questions (Practical Exam Level)

16. Count Blank Lines

Hint:

- Read file line by line.
 - Use `line.strip() == ""`.
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17. Count Digits, Alphabets and Special Characters

Hint:

- Use:
 - `isdigit()`
 - `isalpha()`
 - else → special character
-

18. Display Lines Starting with a Vowel

Hint:

- Read line by line.
 - Check first character after `strip()`.
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19. Count Frequency of a Given Word

Hint:

- Read file.
 - Convert to lowercase.
 - Use `split()`.
 - Count occurrences.
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20. Create Student Result File

Question: Store Name, Physics, Chemistry, Maths marks and display topper.

Hint:

- Calculate total marks.
- Track maximum total while reading records.

